of gravity, where by fiat we limit our attention to long wavelengths and low energies. That's what allows us to calculate the gravitational field in the solar system, even in quantum gravity. But if we want a theory of everything, or at least a theory of gravity that is valid at all possible energies, we're stuck. Something dramatic is called for.

The most popular contemporary approach to quantum gravity is *string theory*, which replaces particles by little loops or segments of one-dimensional "string." (Don't ask what the strings are made of—string stuff is what everything else is made of.) The strings themselves are incredibly small, so much so that they appear like particles when we observe them from a distance.

String theory was initially proposed to help understand the strong nuclear force, but that didn't work out. One of the problems was that the theory inevitably predicts the existence of particles that look and behave exactly like gravitons. That was initially perceived as an annoyance, but pretty soon physicists thought to themselves, "Hmm, gravity actually exists. Maybe string theory is a quantum theory of gravity?" That turns out to be true, and even better there is a bonus: the theory makes finite predictions for all physical quantities, without needing an infinite number of input parameters. The popularity of strings exploded in 1984 when Michael Green and John Schwarz showed that the theory is mathematically consistent.

Today, string theory is the most pursued approach to exploring quantum gravity by a wide margin, although other ideas maintain their adherents. The second-most-popular approach is *loop quantum gravity*, which began as a way of directly quantizing general relativity by using a clever choice of variables—rather than looking at the curvature of spacetime at each point, we consider how vectors are rotated when they travel around closed loops in space. (If space is flat, they don't rotate at all, while if space is curved, they can rotate by a lot.) String theory aspires to be a theory of all the forces and matter at once, while loop quantum gravity only aims at gravity itself. Unfortunately, the obstacles to gathering experimental data relevant to quantum gravity are equally formidable for all the alternatives, so we're stuck not really knowing which approach (if any) is on the right track.

While string theory has been somewhat successful in dealing with the technical problems of quantum gravity, it hasn't shed much light on the conceptual problems. Indeed, one way of thinking about different